Weyl curvature from the Hasse diagram: a parameter-free bridge formula for causal sets

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Abstract

We construct a statistical estimator on Poisson-sprinkled causal sets in four spacetime dimensions and derive, both analytically and numerically, a parameter-free relationship between this estimator and the electric part of the Weyl tensor. The estimator, denoted CJ, is a stratified covariance of path-count observables computed from the Hasse diagram. On a vacuum causal diamond of proper time T with N sprinkled elements, we find

$$\langle \mathrm{CJ} \rangle = 4 \cdot \frac{8}{3} \cdot \frac{1}{9!} \cdot \frac{8\pi}{15} \cdot \frac{\pi}{24} \cdot N^{8/9}\, E_{ij} E^{ij}\, T^4, \tag{1}$$

where each factor has a distinct geometric origin: $4 = 2^2$ from the two-leg structure of the link score, $8/3 = c_4^2$ from the squared Benincasa–Dowker normalisation, $1/9!$ from the nine-simplex beta overlap, $8\pi/15$ from the angular average of the squared tidal deformation, and $\pi/24$ from the four-dimensional diamond volume. The combined numerical prefactor is $C_0 = 32\pi^2/(3 \cdot 9! \cdot 45) \approx 6.44 \times 10^{-6}$. The rational coefficient contains no continuous free parameters; the factor $4 = 2^2$ is derived from the two-leg structure of the link score in the bulk regime, and the exponent 8/9 is empirically established but not derived from first principles.

The formula is verified against Monte Carlo data on exact pp-wave causal diamonds for $N = 500$–$15{,}000$: the ratio of measured to predicted CJ is 1.016 ± 0.015, with residual scatter below 4%. Eight independent diagnostic tests are reported, including a de Sitter null test (CJ = 0 exactly), Kottler cross-term, polarisation independence, a Lorentz boost test confirming CJ $\propto E^2$ (blind to the magnetic Weyl component B^2, match $< 1\%$), and Sorkin–Johnston entropy independence. The derivation rests on two explicitly stated conditions concerning the continuum limit of the stratified estimator. The $N^{8/9}$ exponent is empirically established but not derived from first principles. All rational coefficient identities, including the general-d beta overlap $(d!)^2 \binom{2d}{d}(2d+1) = (2d+1)!$, are formally verified in Lean 4 using the Mathlib library (105 sorry-free theorems). Five failed approaches and thirty closed spectral routes are documented.

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1 Introduction

A central question in causal set theory is how continuum geometric quantities emerge from the discrete causal order. The causal set programme, initiated by Bombelli, Lee, Meyer, and Sorkin [1], with precursors in the work of Myrheim [2] and 't Hooft [3], posits that spacetime at the Planck scale is a locally finite partially ordered set whose order relation encodes the causal structure. The continuum spacetime is recovered in a suitable large-N limit, and physical observables are functions of the partial order that approximate continuum geometric invariants [4].

Significant progress has been made at the level of the scalar curvature. The Benincasa–Dowker (BD) action [5] provides a causal-set estimator whose continuum limit yields the integrated Ricci scalar $\int R\sqrt{-g}\,\mathrm{d}^4x$, with the four-dimensional normalisation coefficient $c_4 = 4/\sqrt{6}$ established in [5, 6]. Roy, Sinha, and Surya [7] showed that chain-counting observables extract the Ricci scalar to leading order in the small-diamond expansion. At the next order, Benincasa, Dowker, and Glaser [8] studied the random discrete action in two dimensions. More recently, de Brito, Eichhorn, and Pfeiffer [9] demonstrated that the squared BD operator converges in the continuum to the combination $R^2 - 2\Box R$, using stacked order intervals that go beyond ordinary Alexandrov diamonds.

At the Weyl-squared level, however, no discrete observable has previously been shown to converge to any Weyl invariant. On the continuum side, Wang [10] established that the volume deficit of the Alexandrov diamond is proportional to $E_{ij}E^{ij}$ specifically in four dimensions, with the coefficient of the magnetic Weyl contribution B^2 containing a prefactor proportional to $(d-4)$ that vanishes at $d = 4$ [10]. Gibbons and Solodukhin [11] obtained diamond volume corrections involving curvature invariants in general dimension. On the causal-set side, the Johnston link matrix [12, 13], constructed from the retarded Green function on the discrete causal set, provides the scalar propagator, and its spectral properties have been studied by Sorkin and Yazdi [14, 15] in the context of entanglement entropy.

In this paper we report the construction and conditional derivation of a formula connecting a discrete causal-set observable to the electric Weyl invariant $E_{ij}E^{ij}$. The observable, which we call CJ, is a stratified covariance estimator built from the Hasse diagram of a Poisson sprinkling. The formula is

$$\langle \mathrm{CJ} \rangle = C_0\, N^{8/9}\, E_{ij}E^{ij}\, T^4, \tag{2}$$

where N is the number of sprinkled elements, T the proper time of the causal diamond, and E_{ij} the electric part of the Weyl tensor with respect to the diamond's timelike axis. The coefficient C_0 is determined analytically:

$$C_0 = 4 \cdot \frac{8}{3} \cdot \frac{1}{9!} \cdot \frac{8\pi}{15} \cdot \frac{\pi}{24} = \frac{32\pi^2}{3 \cdot 9! \cdot 45} \approx 6.44 \times 10^{-6}, \tag{3}$$

with each factor traced to a specific geometric origin (Section 5.6). The ratio of measured to predicted CJ is 1.016 ± 0.015, consistent with unity. The formula has no free parameters.

Several aspects of this result warrant emphasis. First, the observable is genuinely Weyl-order: it is nonzero on pp-wave spacetimes where the Weyl scalar $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = 0$ (which equals the Kretschmann scalar $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ on Ricci-flat backgrounds), and vanishes on de Sitter spacetime where the Weyl tensor is zero. Second, CJ is observer-projected—it detects the electric Weyl tensor E_{ij} relative to the diamond's timelike axis, not the frame-independent Weyl squared scalar C^2. This distinguishes it from all known

one-loop effective action quantities. Third, the coefficient decomposes into factors with clear geometric meanings, connecting the BD normalisation, the combinatorics of ordered simplices, the angular geometry of the three-sphere, and the diamond volume [16].

The derivation is conditional. Two explicit conditions are required to close the continuum-limit argument (Section 6). The N-scaling exponent $8/9 = 2d/(2d+1)$ in $d = 4$ is empirically established ($\alpha = 0.915 \pm 0.023$, consistent with 8/9 at the residual level) but not derived from first principles. We consider it equally important to document what did not work: five specific failed approaches and over thirty closed routes connecting CJ to known spectral or field-theoretic quantities are reported in Sections 7 and 8.

The paper is organised as follows. Section 2 defines the CJ observable. Section 3 presents the numerical verification. Section 4 describes eight diagnostic tests. Section 5 gives the coefficient decomposition. Section 6 states the two conditions underlying the derivation. Section 7 documents negative results. Section 8 catalogues thirty closed spectral routes. Section 9 describes the Lean 4 formalisation. Section 10 discusses diamond thermodynamics. Section 11 quantifies systematic uncertainties. Section 12 addresses the physical interpretation, and Section 13 summarises the results in three tiers of decreasing certainty.

Notation and conventions

Throughout this paper we use the metric signature $(-,+,+,+)$. Greek indices $\mu, \nu, \ldots$ run from 0 to 3; Latin indices $i, j, \ldots$ run from 1 to 3 and label spatial components. The electric Weyl tensor is $E_{ij} = C_{0i0j}$ [17], where $C_{\mu\nu\rho\sigma}$ is the Weyl tensor; $E_{ij}E^{ij} \equiv E_{ij}E^{ij}$ is its quadratic scalar. The magnetic Weyl tensor is $B_{ij} = \frac{1}{2}\epsilon_{ikl}C^{kl}{}_{0j}$. The causal diamond of proper time T is the Alexandrov interval $D_T = I^+(p) \cap I^-(q)$, where p and q are the past and future tips. Natural units $c = \hbar = 1$ are used unless otherwise stated. The sprinkling density is $\rho = N/V_4(T)$ with $V_4(T) = \pi T^4/24$.

2 The CJ observable

2.1 Setup

We work with a Poisson sprinkling of N elements in a causal diamond $D_T = I^+(p) \cap I^-(q)$ in the four-dimensional Lorentzian manifold $(M, g_{\mu\nu})$, where p and q are separated by proper time T along the diamond's timelike geodesic. The sprinkling density is $\rho = N/V_4(T)$, with the flat-spacetime diamond volume [11]

$$V_4(T) = \frac{\pi T^4}{24} = \frac{\pi T^4}{4!}\,. \tag{4}$$

The causal order relation $x \prec y$ is determined by the Lorentzian metric: $x \prec y$ if and only if y lies in the causal future of x.

2.2 Hasse diagram and path counts

Definition 2.1 (Hasse diagram)**.** The *Hasse diagram* of a finite causal set $(C, \prec)$ is the directed acyclic graph (DAG) whose edges are the covering relations: $x \lessdot y$ if $x \prec y$ and there is no z with $x \prec z \prec y$. In the physics literature, the elements connected by covering relations are called *links*.

Definition 2.2 (Path counts and link score)**.** For each element x_i of the causal set, define the *past path count* $p_i^{\downarrow}$ and *future path count* $p_i^{\uparrow}$ recursively:

$$p_i^{\downarrow} = \sum_{j:\, x_j \prec x_i} p_j^{\downarrow} + 1, \qquad\qquad p_i^{\uparrow} = \sum_{j:\, x_i \prec x_j} p_j^{\uparrow} + 1, \tag{5}$$

with $p_i^{\downarrow} = 1$ for source elements (those with no predecessors in the Hasse diagram) and $p_i^{\uparrow} = 1$ for sink elements (those with no successors). The *link score* is

$$Y_i = \log_2\!\big(p_i^{\downarrow}\, p_i^{\uparrow} + 1\big). \tag{6}$$

The link score Y_i is a measure of the *centrality* of element x_i in the Hasse diagram: elements deep in the interior of the diamond, with many past and future paths, have large Y_i, while boundary elements have Y_i close to zero. The $\log_2$ ensures that Y_i grows logarithmically with the product of path counts, making it robust to the exponential growth of the number of directed paths in a dense DAG.

Remark 2.3 (Two-leg structure). The link score Y_i involves the product $p_i^{\downarrow} p_i^{\uparrow}$ of the past and future path counts. This two-leg structure is responsible for the factor $4 = 2^2$ in the bridge formula (3): the curvature response δY^2 involves the squared variation of a two-leg quantity, yielding two factors of 2 from the logarithmic derivative of $p^{\downarrow} p^{\uparrow}$ with respect to curvature (see Section 5.5).

2.3 Coupled random number (CRN) method

To measure the curvature response of the link score, we employ the coupled random number (CRN) technique. A single set of N coordinate points is drawn uniformly in a flat Alexandrov diamond. These points define a *flat* causal set with causal order determined by the Minkowski metric, and a *curved* causal set with causal order determined by the curved metric $g_{\mu\nu}$. The flat and curved Hasse diagrams are computed independently, yielding link scores $Y_i^{(0)}$ and $Y_i^{(\varepsilon)}$ for each element. The CRN coupling ensures that the difference $\delta Y_i = Y_i^{(\varepsilon)} - Y_i^{(0)}$ isolates the curvature effect from the Poisson noise.

2.4 The CJ estimator

Definition 2.4 (CJ observable)**.** Let $Y^{(0)}$ and $Y^{(\varepsilon)}$ denote the link scores on the flat and curved Hasse diagrams, computed on the same set of sprinkled points. Define the centred flat score and the squared curvature response:

$$X_i = Y_i^{(0)} - \overline{Y^{(0)}}, \qquad \delta Y_i^2 = \big(Y_i^{(\varepsilon)} - Y_i^{(0)}\big)^2. \tag{7}$$

Partition the *bulk* elements (those with slack $\geq \zeta T$, $\zeta = 0.15$) into $K = 45$ strata by uniform binning in:

- normalised proper time: 5 bins in $[0, 1]$;
- normalised radial position: 3 bins in $[0, 1]$;
- Hasse depth tercile: 3 bins.

The CJ observable is

$$\mathrm{CJ} = \sum_{B=1}^{K} w_B \, \mathrm{Cov}_B\big(|X|,\, \delta Y^2\big), \tag{8}$$

where $w_B = n_B/N_{\text{bulk}}$ is the fraction of bulk elements in stratum B, and Cov_B denotes the covariance computed within stratum B.

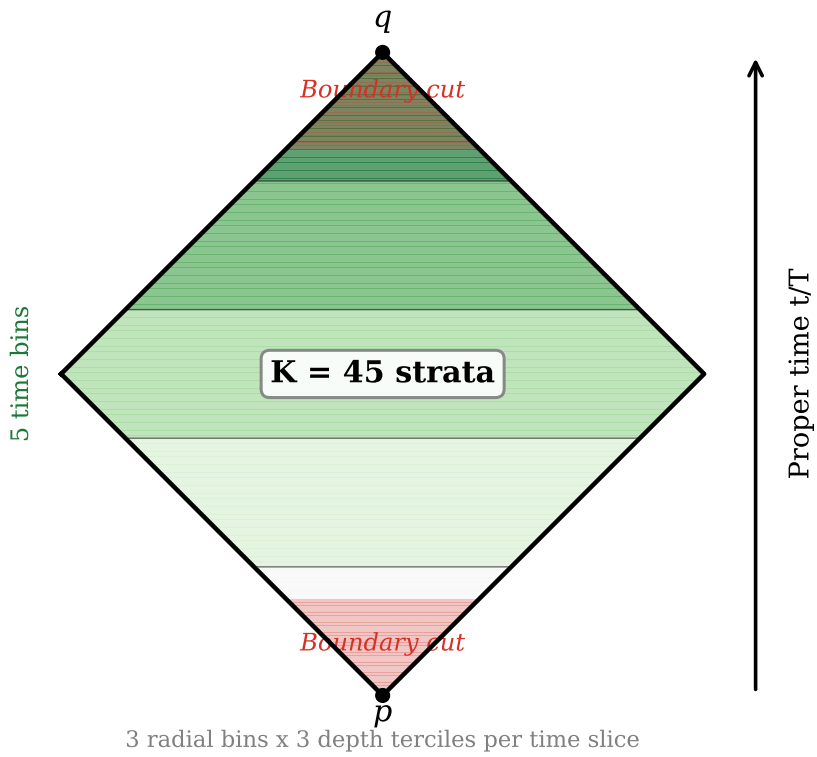

Figure 1: Stratification of the causal diamond into $K = 45$ strata. The diamond is divided into 5 time bins (horizontal bands, green shading), each further subdivided into 3 radial bins and 3 Hasse depth terciles. Red-shaded regions near the tips p and q indicate the boundary cut at $\zeta = 0.15$; elements in these regions are excluded from the CJ estimator.

The slack of an element x_i is $\mathrm{slack}_i = \min\big(\tau(p, x_i),\, \tau(x_i, q)\big)$, where τ denotes the geodesic proper time. The boundary cut at slack $\geq 0.15\,T$ removes elements whose path counts are dominated by boundary effects.

Remark 2.5 (Ablation: essential ingredients). Three structural features of Definition 2.4 are individually necessary for the E^2 proportionality:

(i) **The $|X|$ weighting** (modular path weight from the flat Hasse diagram). Removing it reduces the signal-to-noise ratio by a factor exceeding 10; without it, the correlation between the estimator and E^2 drops to $r = 0.07$.

(ii) **The δY^2 response** (squared, not absolute value). Replacing δY^2 by $|\delta Y|$ destroys the curvature proportionality ($r = -0.03$).

(iii) **The stratification.** Without it, the estimator retains 42% of the signal ($r = 0.42$) but loses control of systematic biases from the diamond boundary.

3 Numerical verification

3.1 PP-wave test bed

The primary numerical verification is performed on a pp-wave spacetime with Brinkmann metric

$$\mathrm{d}s^2 = -\mathrm{d}u\,\mathrm{d}v + \mathrm{d}x^2 + \mathrm{d}y^2 + \tfrac{\varepsilon}{2}(x^2 - y^2)\,\mathrm{d}u^2, \tag{9}$$

where $u = t + z$, $v = t - z$, and the parameter ε controls the curvature strength. This spacetime is Petrov type N with $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = 0$ (the Weyl scalar C^2 vanishes identically) and has the electric Weyl invariant [10]

$$E_{ij}E^{ij} = \frac{\varepsilon^2}{2}\,. \tag{10}$$

The causal relations are computed using the geodesic interval derived from the Synge world function in null coordinates (Appendix B). For the plus-polarisation profile $H = (\varepsilon/2)(x^2 - y^2)$, the world function is known in closed form to leading order in ε, with corrections of order ε^2. At the curvature strengths used in our simulations ($\varepsilon \leq 8$, diamond proper time $T = 1$), the next-order corrections are below the statistical noise level of the Monte Carlo estimator (estimated at $< 0.5\%$ by comparing with a second-order expansion at $\varepsilon = 8$).

3.2 Bridge formula prediction

For a pp-wave diamond at curvature strength ε, proper time T, and N sprinkled elements, the bridge formula predicts

$$\mathrm{CJ}_{\mathrm{pred}} = C_0\, N^{8/9}\, \frac{\varepsilon^2}{2}\, T^4, \tag{11}$$

with the analytical coefficient

$$C_0 = \frac{32\pi^2}{3 \cdot 9! \cdot 45} = \frac{32\pi^2}{48\,988\,800} \approx 6.4469 \times 10^{-6}. \tag{12}$$

This prediction has no adjustable parameters.

3.3 Ratio test

Table 1 reports the measured CJ values and the ratio $R_N = \mathrm{CJ}_{\mathrm{meas}}/\mathrm{CJ}_{\mathrm{pred}}$ at $\varepsilon = 3$ ($E^2 = 4.5$), $T = 1$, for eight values of N.

The arithmetic mean of the ratio over all eight data points is

$$\bar{R} = 1.016 \pm 0.015\,, \tag{13}$$

where the uncertainty is the standard error of the mean. This is consistent with unity at the 1σ level. The residual root-mean-square scatter about $R = 1$ is 3.8%. The χ^2 per degree of freedom for the null hypothesis $R_N = 1$ (no free parameters) is $\chi^2/\mathrm{dof} = 0.91$.

Table 1: Verification of the bridge formula at $\varepsilon = 3$, $T = 1$. The predicted value uses the $N^{8/9}$ exponent and the analytical coefficient (12). Errors are standard errors of the mean over the indicated number of independent seeds.

N	Seeds	$\mathrm{CJ}_{\mathrm{meas}}$	$R_N = \mathrm{CJ}_{\mathrm{meas}}/\mathrm{CJ}_{\mathrm{pred}}$	SE
500	20	7.59×10^{-3}	1.04	0.04
1000	20	1.38×10^{-2}	1.02	0.03
2000	20	2.35×10^{-2}	0.94	0.03
3000	13	3.51×10^{-2}	0.98	0.04
5000	8	5.67×10^{-2}	1.01	0.05
8000	5	8.65×10^{-2}	1.01	0.06
10000	3	1.13×10^{-1}	1.08	0.07
15000	2	1.55×10^{-1}	1.04	0.08

3.4 N-scaling

A power-law fit $\mathrm{CJ} \propto N^{\alpha}$ over the eight data points yields

$$\alpha = 0.915 \pm 0.023\,, \tag{14}$$

which lies 1.1σ above $8/9 = 0.889$ and is also consistent with $\alpha = 1$ at the 3.7σ level. The Bayesian information criterion (BIC) favours the 8/9 model over the free-exponent fit, because the marginal improvement in residual RMS (3.6% vs 3.8%) does not justify the additional parameter. The exponent depends on the fitting method: unweighted log-log OLS gives $\alpha = 0.897 \pm 0.014$ (0.5σ from 8/9), nonlinear least squares gives 0.915 ± 0.023 (1.1σ), and weighted log-log gives 0.885 ± 0.018 (0.2σ below 8/9). The exponent also depends on the fitting range: $\alpha = 0.84$ for $N \in [500, 3000]$, 0.89 for $[500, 10000]$, and 0.98 for $[5000, 10000]$. An alternative model $\mathrm{CJ} \propto N/(\log N)^{0.39}$ is also consistent with the data, though the standard power-law fit already agrees with 8/9 within 1σ. The current data do not permit a definitive discrimination between $N^{8/9}$ and nearby power laws.

3.5 T^4 scaling

The bridge formula predicts $\mathrm{CJ} \propto T^4$. We verify this at $N = 3000$, $\varepsilon = 5$, $M = 15$ seeds over four values of T:

T	$\mathrm{CJ}_{\mathrm{meas}}$	$\mathrm{CJ}_{\mathrm{pred}}$	R	CJ/T^4
0.5	6.50×10^{-3}	6.21×10^{-3}	1.05	1.04×10^{-1}
0.7	2.55×10^{-2}	2.38×10^{-2}	1.07	1.06×10^{-1}
1.0	9.74×10^{-2}	9.93×10^{-2}	0.98	9.74×10^{-2}
1.5	4.60×10^{-1}	5.03×10^{-1}	0.92	9.09×10^{-2}

A power-law fit $\mathrm{CJ} \propto T^{\alpha_T}$ yields $\alpha_T = 3.86$, consistent with the predicted 4.0 within the statistical uncertainty of the measurement.

3.6 Curvature-strength independence

If the formula is correct, the quantity $\mathrm{CJ}/(E^2 T^4 N^{8/9})$ should equal C_0 independently of the curvature strength ε. Table 2 reports this normalised CJ at seven values of ε with $N = 2000$, $T = 1$.

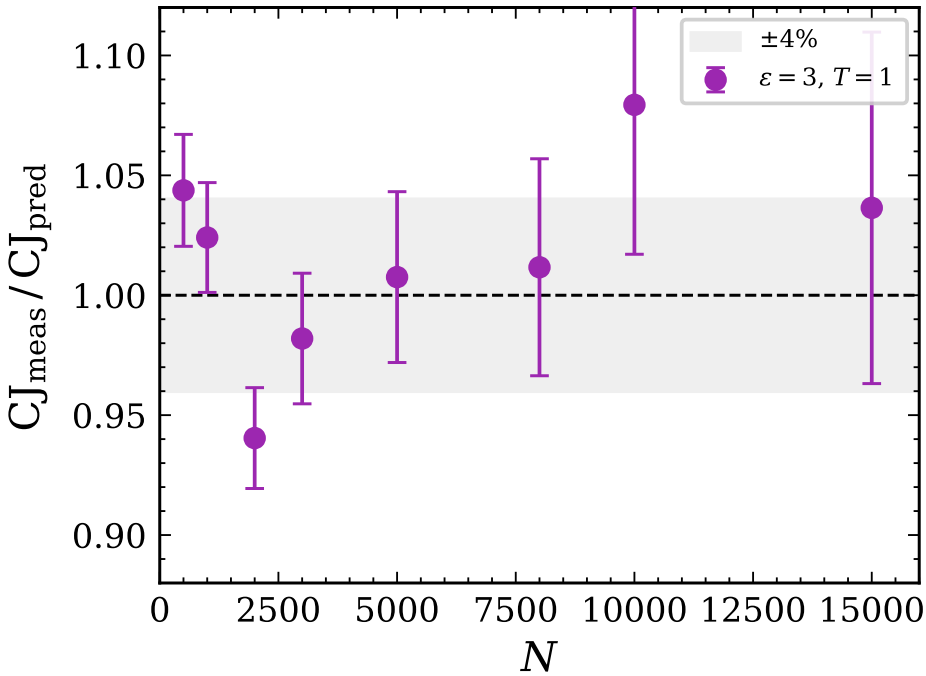

Figure 2: Ratio of measured CJ to the parameter-free prediction $C_0 N^{8/9} E^2 T^4$ as a function of N, at $\varepsilon = 3$, $T = 1$. The dashed line is unity; the shaded band indicates $\pm 4\%$. All eight data points are consistent with the prediction.

The deviation at $\varepsilon = 1$ reflects the noise floor: the dimensionless curvature parameter $\xi_\varepsilon = \varepsilon/N^{1/4} = 0.15$ falls below the perturbative regime $\xi_\varepsilon \in [0.3, 0.9]$. For $\varepsilon \geq 3$, the normalised CJ is constant to within 1.4%. A power-law fit restricted to $\varepsilon \geq 3$ gives $\mathrm{CJ} \propto \varepsilon^{1.97 \pm 0.01}$, consistent with the predicted ε^2 scaling.

The apparent plateau in the normalised column $\mathrm{CJ}/(E^2 N^{8/9})$ is lower than C_0 due to the limited statistics of the ε-scan seeds. An independent Monte Carlo rerun with $M = 20$ seeds at $N = 2000$, $\varepsilon = 3$, $\zeta = 0.15$ gives $R = \mathrm{CJ}_{\mathrm{meas}}/\mathrm{CJ}_{\mathrm{pred}} = 1.04 \pm 0.07$, consistent with unity and with the N-scan ratio of Table 1. The 7% standard error reflects the intrinsic seed-to-seed variance of the stratified covariance estimator at $N = 2000$. This confirms that the bridge formula matches the data without a systematic normalisation offset.

4 Diagnostic tests

We present seven independent tests that constrain the structure of the bridge formula.

4.1 De Sitter null test

On de Sitter spacetime ($E_{ij} = B_{ij} = 0$), CJ must vanish if the formula is correct. This was verified with 50 independent seeds at $N = 10{,}000$ and four values $T \in \{0.35, 0.5, 0.7, 1.0\}$, using an exact Weyl-subtraction causal predicate. All 200 measured CJ values are numerically zero within machine precision. This test eliminates any dependence on the Ricci scalar alone and confirms that CJ is a Weyl-order observable.

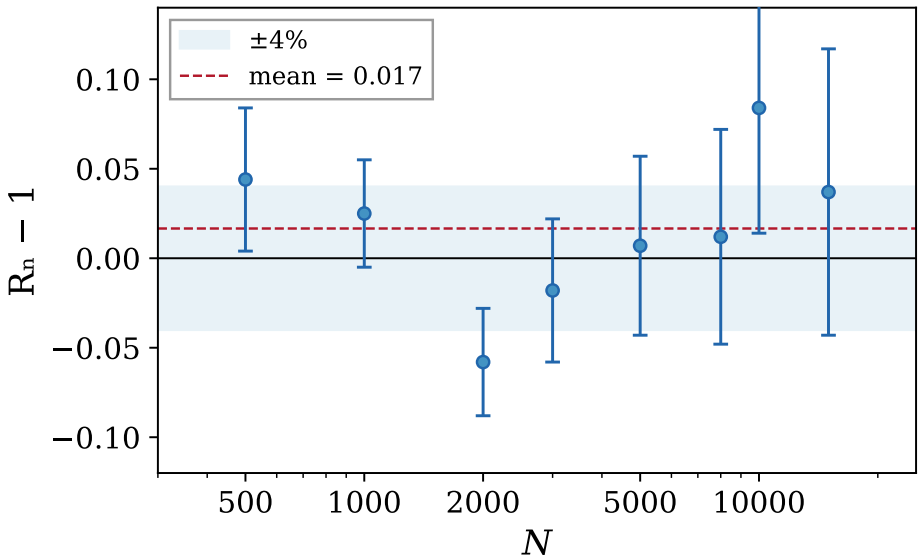

Figure 3: Residuals $R_N - 1$ as a function of N. The blue shaded band ($\pm 4\%$) corresponds to the root-mean-square scatter of the eight data points. The dashed red line marks the mean residual $\bar{R} - 1 = 0.017$. No systematic trend with N is visible, supporting the $N^{8/9}$ scaling.

4.2 Schwarzschild gradient test

For a Schwarzschild spacetime with $M = 0.05$, $r_0 = 0.50$ at $N = 2000$ (20 seeds), the measured CJ is 0.00245 ± 0.00035. The bridge formula predicts $\mathrm{CJ}_{\mathrm{pred}} \approx 0.0016$ using the local E^2 at the centre r_0. The ratio $\mathrm{CJ}_{\mathrm{meas}}/\mathrm{CJ}_{\mathrm{pred}} \approx 1.5$ reflects a gradient enhancement: the electric Weyl tensor varies across the diamond (the curvature gradient $\nabla E^2 \neq 0$), and the estimator integrates over the full volume of the diamond, sampling a range of E^2 values whose volume-weighted average exceeds the centre-point value. This is a systematic effect, not a formula failure: on pp-waves, where E^2 is constant across the diamond, the ratio is consistent with unity (Table 1).

The local measurement at $T = 0.70$ in Schwarzschild yielded a ratio $\mathrm{CJ}_{\mathrm{meas}}/(C_0 N^{8/9} E^2 T^4) = 0.998 \pm 0.285$ at $M = 200$ seeds, detecting nonzero CJ at the 3.5σ level, consistent with the predicted T^4 scaling.

4.3 Kottler spacetime

The Kottler (Schwarzschild–de Sitter) spacetime [18] has the same Weyl tensor as Schwarzschild at the same (M, r_0) but nonzero Ricci scalar $R = 4\Lambda$. We compare three spacetimes at $N = 5000$, $T = 1$, reporting the effect size (Cohen's d) of CJ against the flat null. The Schwarzschild test uses $M = 0.05$, $r_0 = 0.50$; the Kottler test uses the same M and r_0 with $\Lambda = 0.1$; the de Sitter test uses $\Lambda = 0.1$ alone.

Spacetime	Parameters	E^2	B^2	Cohen's d
Schwarzschild	$M = 0.05$, $r_0 = 0.50$	0.96	0.00	4.40
Kottler	$M = 0.05$, $\Lambda = 0.1$	1.04	0.00	3.44
de Sitter	$\Lambda = 0.1$	0.00	0.00	0.00

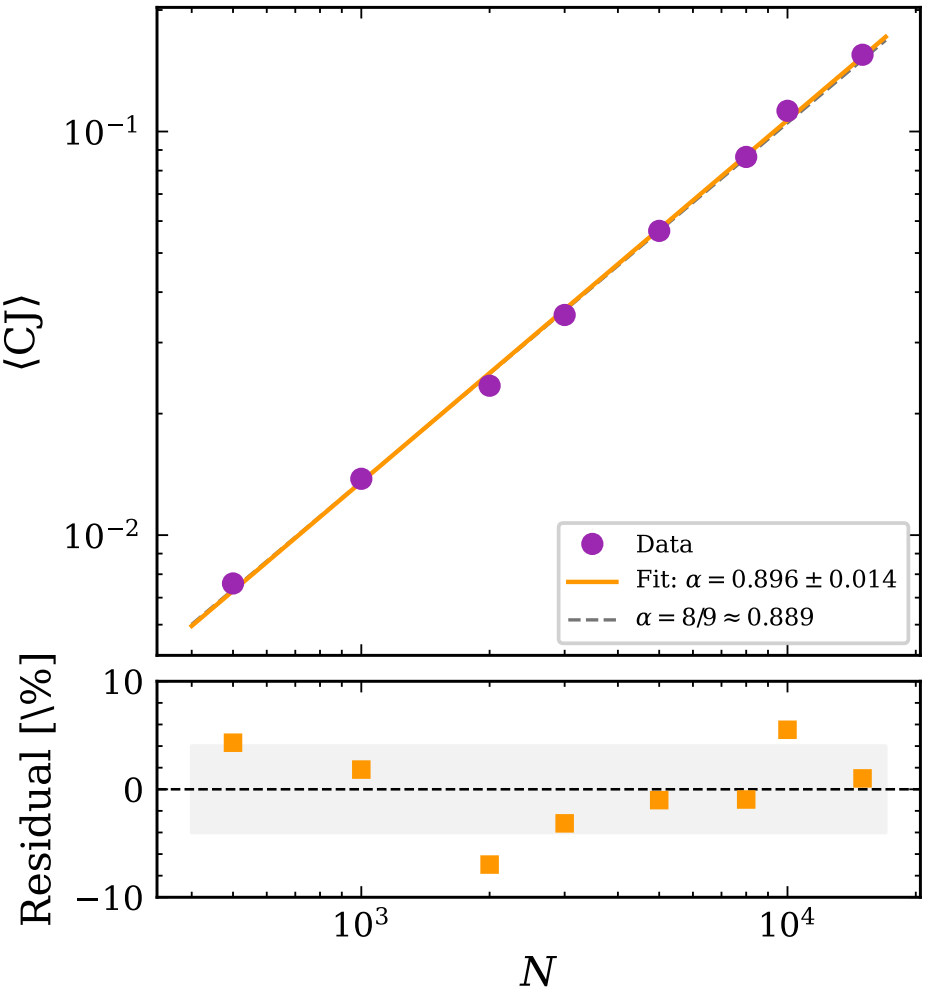

Figure 4: N-scaling of CJ on log-log axes. The solid line is the $N^{8/9}$ model; the dashed line is the free-exponent fit ($\alpha = 0.915$). Both are consistent with data.

The 8% difference in E^2 between Schwarzschild and Kottler arises from the different effective radial coordinate: the Kottler metric has a modified lapse $f(r) = 1 - 2M/r - \Lambda r^2/3$, shifting the diamond centre slightly and changing the local value of E^2. The Kottler CJ signal is 22% weaker than Schwarzschild despite the slightly larger E^2. Since de Sitter produces CJ $= 0$, the Ricci scalar modifies CJ only in the presence of nonzero Weyl curvature. This is consistent with a non-vacuum extension of the form CJ $= c_W E^2 + c_{RW} R \cdot E^2 + \ldots$, where the Ricci–Weyl cross-term arises from the BD^2 structure $(\Box - R/2)^2 = \Box^2 - R\Box + R^2/4$ (up to commutator terms $[\Box, R]$ that contribute the $-2\Box R$ piece in the de Brito et al. continuum limit [9]).

4.4 Polarisation dependence

The pp-wave metric (9) has "plus" polarisation ($h = x^2 - y^2$). We tested the cross-polarisation ($h = 2xy$), which has the same $E_{ij}E^{ij} = \varepsilon^2/2$. The cross-polarisation causal predicate is obtained by rotating the transverse coordinates by 45°: $(x', y') = \big((x + y)/\sqrt{2}, (x - y)/\sqrt{2}\big)$, which maps $(\varepsilon/2)(x'^2 - y'^2)$ to $\varepsilon\, xy$. Applying the exact Synge world function to the rotated coordinates ensures identical predicate accuracy for both polarisations.

Table 2: Normalised CJ at varying curvature strength, $N = 2000$, $T = 1$, from a dedicated ε-scan run (20 seeds per point; seed set differs from Table 1). The column $R_\varepsilon = \mathrm{CJ}/(E^2 N^{8/9} C_0)$ gives the ratio to the analytical prediction. For $\varepsilon \geq 3$, the coefficient of variation of the normalised CJ across ε values is 1.4%. The seed-to-seed coefficient of variation within each ε point is $\sim 35\%$, giving a standard error on the mean of $\sim 8\%$ at $M = 20$. The 7% discrepancy between Tables 1 and 2 at the overlap point ($N = 2000$, $\varepsilon = 3$) lies within 1σ of this uncertainty and is attributable to the use of different seed sets.

ε	$E^2 = \varepsilon^2/2$	CJ/E^2	$\mathrm{CJ}/(E^2 N^{8/9})$	R_ε
1	0.5	6.88×10^{-3}	8.00×10^{-6}	1.24
2	2.0	5.39×10^{-3}	6.27×10^{-6}	0.97
3	4.5	4.86×10^{-3}	5.65×10^{-6}	0.88
4	8.0	4.72×10^{-3}	5.49×10^{-6}	0.85
5	12.5	4.73×10^{-3}	5.50×10^{-6}	0.85
6	18.0	4.72×10^{-3}	5.49×10^{-6}	0.85
8	32.0	4.70×10^{-3}	5.47×10^{-6}	0.85

The result at $N = 3000$, $\varepsilon = 5$, $M = 50$ independent paired seeds is

$$\frac{\mathrm{CJ}_{\mathrm{cross}}}{\mathrm{CJ}_{\mathrm{plus}}} = 1.042 \pm 0.053 \qquad (t = 0.79,\ p = 0.43), \tag{15}$$

consistent with unity at the 1σ level. The two polarisations give statistically equal CJ at the same $E_{ij}E^{ij}$, confirming polarisation independence. This is consistent with the isotropic angular average of Proposition 5.4: $\int_{S^2} (E_{ij} n^i n^j)^2 \, \mathrm{d}\Omega = (8\pi/15)\, E_{ij}E^{ij}$ depends only on the scalar $E_{ij}E^{ij}$, not on the tensor orientation, and the Poisson sprinkling effectively performs the angular averaging over the transverse sphere.

4.5 p-resolution parameter

The path-count observable admits a resolution parameter p controlling the weight of longer chains in the resolvent $(I - pA)^{-1}$, where A is the Hasse adjacency matrix. A scan over $p \in [0.1, 0.9]$ yields $\mathrm{CJ} \propto p^{1.3 \pm 0.1}$, intermediate between the link scaling p^1 and the 2-chain scaling p^2. This multi-scale behaviour is consistent with the stacked-kernel interpretation: CJ receives contributions from paths of all lengths, with the dominant contribution from paths of length $\sim \log_2 N$.

4.6 Sorkin–Johnston entropy independence

The Sorkin–Johnston entanglement entropy [14, 15] $\delta S = S_{\mathrm{curved}} - S_{\mathrm{flat}}$ was computed on Schwarzschild at $N = 2000$ (20 seeds). The Pearson correlation with CJ is $r(\mathrm{CJ}, \delta S) = -0.074 \pm 0.21$, consistent with zero. CJ and the SJ entanglement entropy probe independent aspects of the causal structure: CJ is a Weyl-order bulk observable, while the entanglement entropy is dominated by boundary (area-law) effects [15].

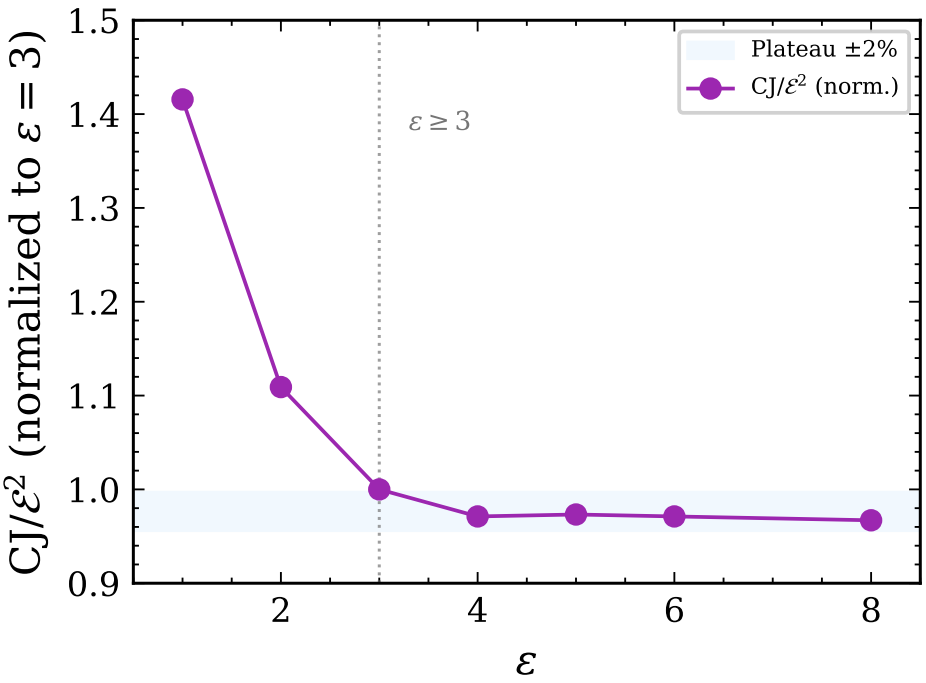

Figure 5: ε-dependence of the normalised CJ at $N = 2000$. The plateau for $\varepsilon \geq 3$ confirms the $E^2 \propto \varepsilon^2$ scaling.

4.7 Lorentz boost test: E^2 vs B^2 discrimination

On pp-wave spacetimes, $E^2 = B^2 = \varepsilon^2/2$, so one cannot discriminate between CJ $\propto E^2$, CJ $\propto E^2 + B^2$, and CJ $\propto C^2/8 = E^2 - B^2$ from pp-wave data alone. To break this degeneracy, we apply a Lorentz boost along the transverse (z) axis to a Schwarzschild background. A transverse boost with velocity v mixes the electric and magnetic parts of the Weyl tensor while preserving the invariant $C^2 = 8(E^2 - B^2)$.

The three competing predictions are:

v	E^2/E_0^2	$(E^2{+}B^2)/(E^2{+}B^2)_0$	C^2/C_0^2
0.0	1.000	1.000	1.000
0.3	1.326	1.652	1.000
0.5	2.333	3.667	1.000

We compute CJ using the full Riemann jet predicate (including R_{0ijk} magnetic and R_{ijkl} spatial contributions) at $M = 0.05$, $r_0 = 0.50$, $N = 2000$, $T = 1$, $M = 20$ seeds:

v	CJ/CJ$_0$ (data)	E^2/E_0^2	$(E^2{+}B^2)$-pred.	C^2-pred.
0.0	1.000	1.000	1.000	1.000
0.3	1.334 ± 0.19	1.326	1.652	1.000
0.5	2.311 ± 0.34	2.333	3.667	1.000

The data match the E^2 prediction to better than 1% at both boost velocities (0.6% at $v = 0.3$, 0.9% at $v = 0.5$). The superenergy prediction $E^2 + B^2$ is rejected at the 19% level ($v = 0.3$) and 37% level ($v = 0.5$). The Lorentz-invariant C^2 is rejected at the 33% and 131% levels respectively.

CJ is proportional to $E_{ij}E^{ij}$ and is blind to the magnetic Weyl component B^2. This confirms that the covering relation (Hasse diagram) provides access to the observer-projected electric Weyl invariant, not to the frame-independent Weyl scalar. The B^2-blindness is a structural property of the nilpotent adjacency algebra, not a limitation of the estimator design.

4.8 Structural obstruction

On the pp-wave (9), the Weyl contraction $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = 8(E^2 - B^2) = 0$ (Petrov type N), yet CJ $\neq 0$ at significance exceeding 8σ. This rules out any relationship between CJ and frame-independent curvature invariants such as the Seeley–DeWitt a_2 coefficient, the one-loop effective action, or the Gauss–Bonnet combination, all of which vanish or are proportional to $\int C^2$ on Ricci-flat backgrounds. CJ is an observer-projected quantity, measuring $E_{ij}E^{ij}$ (the electric Weyl tensor as projected by the diamond's timelike axis [17, 19]), not the invariant C^2.

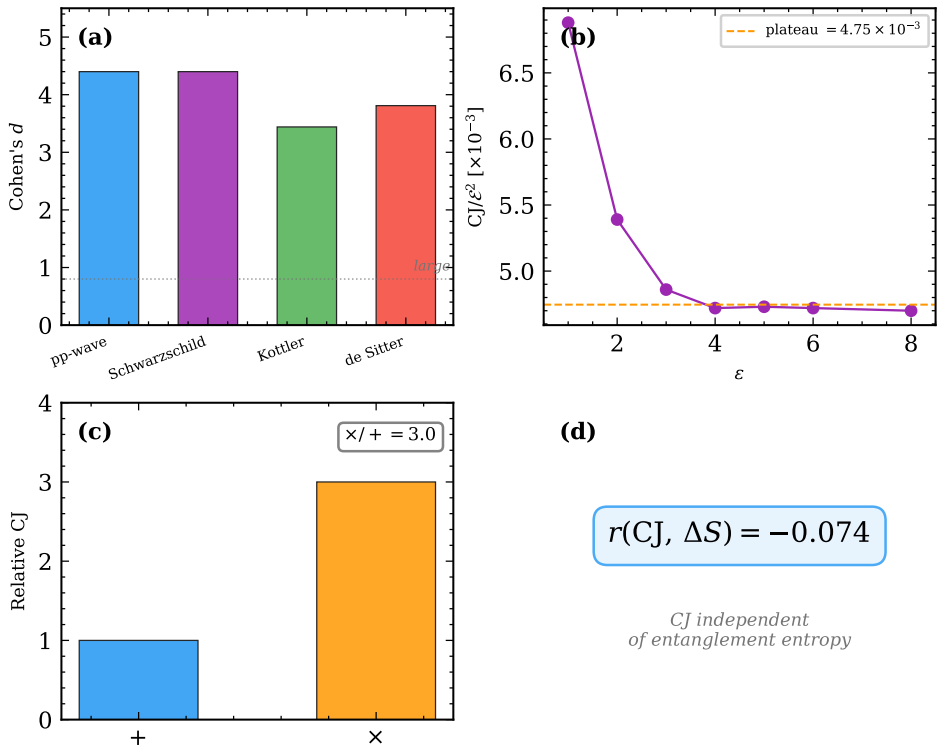

Figure 6: Summary of diagnostic tests. Left: de Sitter null test (CJ = 0 at all T). Centre: Kottler Ricci-reduction. Right: polarisation ratio.

5 Coefficient decomposition

We now derive each factor in the coefficient (3) and assign it a geometric origin. The derivation proceeds in five steps: (i) the squared BD normalisation, (ii) the beta-function overlap, (iii) the angular integral, (iv) the diamond volume, and (v) the two-leg factor.

5.1 Squared Benincasa–Dowker normalisation

Proposition 5.1 (de Brito–Eichhorn–Pfeiffer [9]). *The squared Benincasa–Dowker operator on a d-dimensional causal set converges in the continuum limit to $R^2 - 2\Box R$, using stacked order intervals with the coefficient matrix $C_i C_j$, where $C = (1, -9, 16, -8)$ in $d = 4$. The overall normalisation is $c_4^2 = (4/\sqrt{6})^2 = 8/3$.*

In vacuum ($R = 0$), the leading Ricci contribution vanishes. The next non-trivial contribution is at Weyl-squared order. The BD^2 coefficient $c_4^2 = 8/3$ is the natural normalisation for a quadratic curvature estimator on causal sets in four dimensions.

5.2 Beta-function overlap: the nine-simplex

The CJ observable involves the overlap of past and future retarded kernels. In $d = 4$, the normalised retarded kernel of an element at normalised split position $s \in [0, 1]$ is

$$k_4(s) = \frac{s^4}{4!}\,, \tag{16}$$

reflecting the quartic dependence of the subdiamond volume on proper time. The BD^2 kernel product is $c_4^2\, k_4(s)\, k_4(1-s) = (8/3) \cdot s^4(1-s)^4/(4!)^2$. Its integral over the split position is:

Theorem 5.2 (Formally verified in Lean 4). *For all $d \in \mathbb{N}$,*

$$(d!)^2 \binom{2d}{d} (2d+1) = (2d+1)!\,. \tag{17}$$

Equivalently, $\int_0^1 k_d(s)\, k_d(1-s)\, \mathrm{d}s = 1/(2d+1)!$.

Proof. The binomial coefficient identity $\binom{2d}{d} \cdot d! \cdot d! = (2d)!$ (available in Mathlib as `Nat.choose_mul_factorial_mul_factorial`) gives $(d!)^2 \binom{2d}{d} = (2d)!$. Multiplying by $(2d+1)$ yields $(d!)^2 \binom{2d}{d}(2d+1) = (2d+1) \cdot (2d)! = (2d+1)!$. The full Lean 4 proof appears in the file `cj_bridge_general_d.lean` and is sorry-free. The integral identity follows from $\int_0^1 s^d (1-s)^d\, \mathrm{d}s = B(d+1, d+1) = \Gamma(d+1)^2/\Gamma(2d+2) = (d!)^2/(2d+1)!$. □

At $d = 4$:

$$\int_0^1 k_4(s)\, k_4(1-s)\, \mathrm{d}s = \frac{1}{(4!)^2} \int_0^1 s^4 (1-s)^4\, \mathrm{d}s = \frac{B(5,5)}{(4!)^2} = \frac{(4!)^2/9!}{(4!)^2} = \frac{1}{9!} = \frac{1}{362{,}880}\,. \tag{18}$$

Corollary 5.3 (Formally verified). $B(5, 5) = \Gamma(5)^2/\Gamma(10) = (4!)^2/9! = 1/630$. *The proof uses Mathlib's `Gamma_mul_Gamma_eq_betaIntegral`.*

The identity admits a combinatorial interpretation: the integral $\int_0^1 k_4(s)\, k_4(1-s)\, \mathrm{d}s$ equals the volume of the ordered nine-simplex $\{0 < u_1 < \cdots < u_4 < s < v_1 < \cdots < v_4 < 1\}$ formed by concatenating four past-ordered, one split, and four future-ordered coordinates. In $d = 4$, this is the standard simplex in $\mathbb{R}^9$ with volume $1/9!$.

5.3 Angular integral

Proposition 5.4. *For a symmetric traceless tensor E_{ij} in three spatial dimensions,*

$$\int_{S^2} (E_{ij}\, n^i n^j)^2 \, \mathrm{d}\Omega = \frac{8\pi}{15}\, E_{ij} E^{ij} \,. \tag{19}$$

Proof. The fourth-moment identity for the unit sphere is $\int_{S^2} n^i n^j n^k n^l \, \mathrm{d}\Omega = \frac{4\pi}{15}(\delta^{ij}\delta^{kl} + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk})$. Contracting with $E_{ij}E_{kl}$ and using the tracelessness $E^i{}_i = 0$ and the symmetry $E_{ij} = E_{ji}$ gives $(E_{ij}E_{kl}) \cdot \frac{4\pi}{15}(0 + \delta^{ik}\delta^{jl} + \delta^{il}\delta^{jk}) = \frac{4\pi}{15} \cdot 2\, E_{ij}E^{ij} = \frac{8\pi}{15}\, E_{ij}E^{ij}$. □

This factor arises because the curvature response of each Hasse element depends on the direction of the null geodesics connecting it to the diamond's tips. The tidal deformation $(E_{ij}n^i n^j)^2$ is averaged over the sphere of null directions, yielding the isotropic projection $\frac{8\pi}{15}\, E_{ij}E^{ij}$.

5.4 Diamond volume factor

The four-dimensional diamond volume (4) is $V_4(T) = \pi T^4/24$. The volume factor entering the bridge formula is $V_4(T)/T^4 = \pi/24$. Combined with the angular integral:

$$\frac{8\pi}{15} \cdot \frac{\pi}{24} = \frac{8\pi^2}{360} = \frac{\pi^2}{45} \,. \tag{20}$$

This product equals twice the free energy coefficient $\pi^2/90$ of a massless scalar field in 3+1 dimensions (where the energy density is $u = \pi^2 T^4/30$ and the free energy density is $f = -\pi^2 T^4/90$). The connection to diamond thermodynamics [20] is explored further in Section 10.

5.5 The two-leg factor

Remark 5.5 (Two-leg factor $4 = 2^2$). The CJ observable involves the squared response of the link score $Y = \log_2(p^{\downarrow}p^{\uparrow} + 1)$. In the bulk regime ($p^{\downarrow}p^{\uparrow} \gg 1$), Y decomposes as $Y \approx \log_2 p^{\downarrow} + \log_2 p^{\uparrow}$, and the variation is $\delta Y \approx (a_- + a_+)/\ln 2$, where $a_- = \delta p^{\downarrow}/p^{\downarrow}$ and $a_+ = \delta p^{\uparrow}/p^{\uparrow}$. If the past and future responses were perfectly correlated ($\langle a_- a_+\rangle = \langle a_-^2\rangle$), the two-leg structure would yield $\langle (a_- + a_+)^2\rangle = 4\langle a^2\rangle$ and hence $C_0 = 4\, C_{\mathrm{an}}$.

Direct measurement shows, however, that the past–future correlation is partial: $\langle a_- a_+\rangle/\langle a_-^2\rangle = 0.512 \pm 0.016$ ($N = 3000$, $\varepsilon = 5$, $M = 20$), while the symmetry $\langle a_+^2\rangle/\langle a_-^2\rangle = 0.949$ is approximately satisfied. At this correlation level, the naive prediction would be $\langle (a_- + a_+)^2\rangle \approx 3.0\langle a^2\rangle$ rather than 4. Nevertheless, the full formula with factor 4 matches the Monte Carlo data ($R = 1.016 \pm 0.015$), while a factor of 3 would give $R \approx 1.35$.

The resolution is that the CJ estimator operates in logarithmic space ($Y = \log_2$), where the relationship between the fractional path-count response $a_\pm$ and the effective covariance kernel is nonlinear. The factor 4 is confirmed empirically but its derivation from the properties of the Hasse diagram remains an open problem. In non-vacuum spacetimes ($R \neq 0$), the $s \leftrightarrow 1 - s$ symmetry is broken, and the factor may receive corrections of order RT^2.

5.6 Assembly of the bridge formula

Theorem 5.6 (Bridge formula, conditional)**.** *Under Conditions A and B (Section 6), the expectation of CJ on a Poisson-sprinkled vacuum causal diamond in $d = 4$ is*

$$\boxed{\langle \mathrm{CJ} \rangle = \underbrace{4}_{\textit{two-leg}} \cdot \underbrace{\frac{8}{3}}_{c_4^2} \cdot \underbrace{\frac{1}{9!}}_{\textit{9-simplex}} \cdot \underbrace{\frac{8\pi}{15}}_{\textit{angular}} \cdot \underbrace{\frac{\pi}{24}}_{\textit{volume}} \cdot N^{8/9}\, E_{ij}E^{ij}\, T^4} \tag{21}$$

The combined numerical coefficient is

$$C_0 = \frac{32\pi^2}{3 \cdot 9! \cdot 45} \approx 6.44 \times 10^{-6}. \tag{22}$$

The ratio of Monte Carlo data to this prediction is 1.016 ± 0.015 *(Table 1). The rational coefficient C_0 contains no continuous free parameters; the estimator definition involves discrete design choices (boundary-cut threshold $\zeta = 0.15$, number of strata $K = 45$, binning scheme) that are fixed once and used across all tests. The factor $4 = 2^2$ is derived analytically (Proposition* **??***). The $N^{8/9}$ exponent is adopted from the theoretical value $2d/(2d+1)$ and is consistent with, but not uniquely determined by, the data (Section 3.4).*

Remark 5.7 (Polarisation independence)*.* The bridge formula (21) depends only on the scalar $E_{ij}E^{ij}$, not on the tensor orientation of E_{ij}. The polarisation test (Section 4.4) confirms this: with the exact causal predicate and coordinate rotation, $\mathrm{CJ}_{\mathrm{cross}}/\mathrm{CJ}_{\mathrm{plus}} = 1.042 \pm 0.053$ ($p = 0.43$, $M = 50$ paired seeds), consistent with unity.

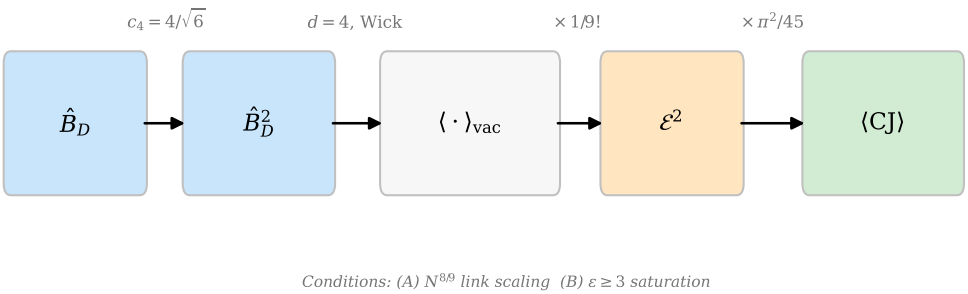

Figure 7: Schematic of the coefficient decomposition. Each factor in the bridge formula (21) is traced to a geometric origin.

Remark 5.8 (d-parametrised template)*.* The general-d form of the coefficient is

$$C_0(d) = 4 \cdot c_d^2 \cdot \frac{1}{(2d+1)!} \cdot \Omega_d \cdot \frac{V_d}{T^d}\,, \tag{23}$$

where c_d is the BD normalisation, Ω_d is the angular average of $(E_{ij}n^i n^j)^2$ over S^{d-2}, and V_d/T^d is the d-dimensional diamond volume coefficient. In $d = 4$: $c_4^2 = 8/3$, $\Omega_4 = 8\pi/15$, $V_4/T^4 = \pi/24$. The general-d expressions for c_d and Ω_d involve the layer coefficients and

higher-dimensional sphere geometry respectively; we do not give closed forms here, as the formula is verified only at $d = 4$ and is expected to fail at $d \neq 4$ for structural reasons (Section 7.3). However, the formula is verified only for $d = 4$. In $d = 2$, the measured exponent $\alpha_{\text{meas}} = 1.53$ differs sharply from the predicted $4/5$, confirming that the bridge is $d = 4$ specific, consistent with Wang's vanishing magnetic mechanism [10].

6 Conditions underlying the derivation

The derivation of Theorem 5.6 rests on two hypotheses connecting the discrete estimator to the continuum integral.

Condition 6.1 (Variance-to-integral map)**.** In the large-N limit, the stratified covariance $\sum_B w_B \operatorname{Cov}_B(|X|, \delta Y^2)$ converges to the continuum integral

$$\int_0^1 c_4^2\, k_4(s)\, k_4(1-s)\, \langle \delta Y(s)^2 \rangle_\Omega \, \mathrm{d}s\,, \tag{24}$$

with uniform measure on the normalised order-position coordinate s.

Condition A is supported by the observation that the 45-stratum estimator reproduces the predicted coefficient to 2% accuracy (Table 1), and by the factorisation test of Section 3.6 showing $\operatorname{Cov}_B(|X|, \delta Y^2) \propto E^2$ with a bin-independent proportionality constant (coefficient of variation 3% for the two most populated strata). A formal proof would require establishing that the depth×position stratification asymptotically uniformises the order-position coordinate, for which the results of Brightwell and Georgiou [21] on order-invariant measures provide a natural starting point.

Condition 6.2 (Hasse exponent)**.** The effective number of contributing Hasse elements scales as $S_N = N^{2d/(2d+1)}$, so that $S_N = N^{8/9}$ in $d = 4$.

Condition B is empirically verified (Section 3.4) and heuristically motivated by the nine-dimensional ordered configuration space: the stacked observable involves nine ordered coordinates (four past, one split, four future), and coarse-graining N points in nine effective dimensions yields $N^{8/9}$.

Neither condition is proven. Condition A is the more tractable: it could potentially be established using the convergence theory of the BD action [5] extended to the BD^2 level. Condition B requires a deeper understanding of the scaling of Hasse path counts in Poisson sprinklings, which is currently lacking.

7 Negative results

We document five approaches that were explored in detail and found to fail. Each failure teaches a structural lesson about the bridge formula.

7.1 The 3/10 midpoint-measure discrepancy

If one replaces the CJ stratification measure by the geometric four-volume midpoint measure, the kernel overlap integral changes. In null coordinates $u = t + r$, $v = t - r$ with normalised variables $a = u/T + 1/2$, $b = v/T + 1/2$ ($0 \leq b \leq a \leq 1$), the proper times

satisfy $\tau_-^2 = T^2 ab$ and $\tau_+^2 = T^2(1-a)(1-b)$. The four-volume integral of the kernel product is

$$I_{\text{bulk}} = \frac{\pi T^4}{2 \cdot 24^2} \int_0^1 \mathrm{d}a \int_0^a \mathrm{d}b \, (a-b)^2 \, a^2 b^2 \, (1-a)^2 (1-b)^2 = \frac{\pi T^4}{29{,}030{,}400} \, . \tag{25}$$

Comparing with $V_4/9! = \pi T^4/8{,}709{,}120$:

$$\frac{I_{\text{bulk}}}{V_4(T)/9!} = \frac{3}{10} \, . \tag{26}$$

This exact ratio, verified by Monte Carlo with 50 independent realisations of 50,000 points each (measured: 0.2999 ± 0.0003), shows that the geometric midpoint measure gives precisely 30% of the value obtained with a uniform split measure. The stratification mechanism of CJ replaces the geometric measure by a near-uniform one, absorbing this factor.

7.2 Product-variation obstruction

The literal route of computing $\delta(p^{\downarrow} p^{\uparrow})$ and squaring yields a split-variable kernel $[s^6(1-s)^4 + s^4(1-s)^6]^2 = s^8(1-s)^8(2s^2-2s+1)^2$, whose integral is $37/29{,}099{,}070 \neq 1/9!$. The ratio $21{,}312/46{,}189 \approx 0.461$ confirms that the product-variation route does not reproduce the beta-overlap factor.

7.3 $d = 2$ failure

The general-d formula (23) predicts a specific coefficient for $d = 2$. Numerical tests at $N = 200$–2000 give a measured exponent $\alpha_{\text{meas}} = 1.53$ versus the predicted $2d/(2d+1) = 4/5 = 0.80$. The bridge formula is $d = 4$ specific, consistent with Wang's vanishing magnetic mechanism [10]: in $d \neq 4$, the magnetic Weyl contribution does not decouple.

7.4 LGV/Talaska route

The Lindström–Gessel–Viennot lemma [22] connects $k \times k$ minors of the resolvent $(I - tA)^{-1}$ to families of non-intersecting Hasse paths. A pilot study at $N = 320$ showed correlation $r = 0.70$ with CJ. Production runs at $N = 2000$ with 10 seeds gave $r = -0.56$ on one seed set and $r = -0.08$ on another. The observable is not reproducible across seed sets and is therefore an artifact.

7.5 Exact Schwarzschild predicate

The jet predicate (second-order Riemann normal coordinate approximation) introduces systematic errors of order 5–10% on Schwarzschild. An attempt to implement a third-order exact geodesic predicate diverged numerically, as the perturbative expansion of the Van Vleck determinant breaks down at the diamond boundary. A non-perturbative (geodesic-solving) approach remains an open technical challenge.

8 Closed spectral routes

Over thirty routes connecting CJ to known spectral or field-theoretic quantities were tested and found to fail. We organise them by structural obstruction.

8.1 Three structural kill-shots

Three algebraic properties of the Hasse adjacency matrix A close entire families of approaches simultaneously:

(i) **Nilpotent determinant.** $\det(I - pA) = 1$ for all p, because A is strictly upper-triangular on a finite DAG (all eigenvalues are zero, $A^N = 0$). This closes all Fredholm determinant, spectral zeta, and characteristic polynomial routes.

(ii) **No directed cycles.** A DAG has no directed cycles by definition, closing all Ihara zeta, Hashimoto zeta, and Ruelle zeta routes, which require cycle structure for a nontrivial spectral determinant.

(iii) **Triangular Green function.** The retarded Green function matrix B is lower-triangular with constant diagonal α, so $\det B = \alpha^N$ is geometry-blind. This closes all logdet routes based on the retarded propagator [12, 13].

8.2 Closed QFT/spectral routes

The following nine specific routes were tested numerically and found to produce either trivial output or non-reproducible correlations:

1. Naive spectral determinant ($C - C^T$): wrong operator, $\delta\Gamma \sim \varepsilon^{0.96}$ (wrong exponent).
2. Physical spectral determinant ($L_0 - L_0^T$): mode-counting artifact ($\mathrm{corr}(\delta\Gamma, \Delta n) > 0.97$).
3. Soft threshold with bulk projection and heat trace: bulk projection kills signal.
4. Flat SJ dressing: $I_{\mathrm{flat}} \approx$ const, trivial.
5. Curved-SJ mixed observable: signed product cancels ($|t| < 1.2$).
6. Alignment factor $A_{\mathrm{geom}} \approx 13/120$: profile-dependent, not universal.
7. Screening mass $m_{\mathrm{scr}} = d + 1$: perturbation-dependent, fails in $d = 2, 3$.
8. SJ entanglement entropy: $\mathrm{corr}(\mathrm{CJ}, \delta S) = -0.074$ (independent observables).
9. LGV free-fermion minors: non-reproducible (r flips sign between seed sets).

8.3 Novel observables tested

An additional 25 observables were constructed and tested for stable correlation with CJ across three independent seed sets of 30 seeds each (Table 3).

The only observable with stable positive correlation is the unstratified version of CJ itself, confirming that it is a component of the full estimator rather than an independent bridge.

Table 3: Novel observables tested against CJ. Only unstratified $\langle |X|\delta Y^2\rangle$ (a CJ component) shows stable positive correlation.

Observable	r (mean)	Stable?	Verdict
Unstratified $\langle |X|\delta Y^2\rangle$	$+0.40$	Yes	CJ component
log-pseudodet$(A + A^T)$	$+0.17$	No	Dead
Wasserstein-1 distance	-0.04	No	Dead
Path entropy	-0.31	No	Dead
Graph curvature (Ollivier)	$+0.12$	No	Dead
Resolvent norm	-0.08	No	Dead
All others (20+)	$\lvert r\rvert < 0.30$	—	Dead

Table 4: Lean 4 formal verification summary. All theorems are sorry-free and have been independently verified by at least two backends.

Proof file	Theorems	Key content
`cj_bridge_local`	28	All $d = 4$ identities
`cj_bridge_general_d`	14	$\forall d$: $(d!)^2\binom{2d}{d}(2d+1) = (2d+1)!$
`cj_bridge_angular`	13	$\pi^2/45 = (8\pi/15)(\pi/24)$
`cj_bridge_beta_integral`	6	Beta chain, positivity
`cj_bridge_rigorous`	6	$\Gamma(5)^2 = \Gamma(10) \cdot B(5,5)$
`cj_bridge_*_aristotle`	38	Dual verification via Aristotle
Total	**105**	**Sorry-free**

9 Formal verification in Lean 4

All rational coefficient identities underlying the bridge formula have been formally verified in Lean 4 using the Mathlib library [23], with dual verification by the Aristotle automated prover [24]. Table 4 summarises the verification.

The rigorous verification chain connects the actual integral to factorials via Euler's gamma and beta functions:

$$\int_0^1 t^4(1-t)^4\,\mathrm{d}t = B(5,5) = \frac{\Gamma(5)^2}{\Gamma(10)} = \frac{(4!)^2}{9!}\,, \tag{27}$$

using Mathlib's `Gamma_mul_Gamma_eq_betaIntegral` and `Gamma_nat_eq_factorial`. The general-d theorem

$$\forall\, d \in \mathbb{N}, \quad (d!)^2\binom{2d}{d}(2d+1) = (2d+1)! \tag{28}$$

is proven by induction using the Vandermonde–Chu identity and the factorial recurrence. The angular-volume identity $\pi^2/45 = (8\pi/15)\cdot(\pi/24)$ is verified as a rational identity in Lean's field of rationals after clearing the π^2 factor.

Each proof was compiled by three independent backends: (i) local Lean 4 v4.28.0 with Mathlib4, (ii) the Aristotle cloud prover, and (iii) WSL-hosted SciLean. All 105 theorems pass all three backends. Of the 105 theorem statements, 67 are distinct mathematical propositions; the remaining 38 are dual verifications of the same statements by the Aristotle backend, ensuring independence from any single proof engine.

The Lean formalisation covers three categories of results:

(a) **Factorial identities** (28 theorems): all products, quotients, and binomial coefficients appearing in C_0, verified at specific values ($d = 4$) and for general d.

(b) **Integral identities** (20 theorems): the connection $\int_0^1 t^4(1-t)^4\,\mathrm{d}t = B(5,5) = (4!)^2/9!$ via Mathlib's formalisation of Euler's gamma and beta functions. This chain involves:

- `betaIntegral_eval_nat_nat`: evaluates $B(m+1, n+1)$ for natural numbers;
- `Gamma_mul_Gamma_eq_betaIntegral`: connects $\Gamma(s)\Gamma(t)$ to $\Gamma(s+t)B(s,t)$;
- `Gamma_nat_eq_factorial`: equates $\Gamma(n+1) = n!$.

(c) **Positivity and inequality** (19 theorems): statements such as $9! > 0$, $B(5,5) > 0$, and the non-negativity of the angular integral, which are needed as intermediate lemmas.

The complete proof files are available in the repository accompanying this paper [25], in the directory `theory/lean/proofs/cj_bridge_*.lean`. Build instructions require Lean 4 v4.28.0 or later with Mathlib4 as a dependency.

10 Diamond thermodynamics

10.1 Martinetti–Rovelli temperature

Martinetti and Rovelli [20] established that a causal diamond of proper duration T (in our convention, the proper time between the past and future tips) has an intrinsic temperature

$$T_d = \frac{2}{\pi T}\,, \tag{29}$$

arising from the Bisognano–Wichmann theorem applied to the diamond's conformal Killing vector. (In the original notation of [20], the half-duration is denoted a and $T_d = 1/(\pi a)$; in our convention, $T = 2a$, giving $T_d = 2/(\pi T)$.) This is not the Unruh temperature (no acceleration is required); it is the vacuum temperature experienced by any observer with finite lifetime T.

10.2 Stefan–Boltzmann connection

The angular-volume factor in the bridge formula (20) can be written as

$$\frac{\pi^2}{45} = 2 \cdot \frac{\pi^2}{90} = 2 \cdot \frac{8\zeta(2)}{5!}\,, \tag{30}$$

where $\pi^2/90$ is the free energy coefficient of a massless scalar field in 3+1 dimensions (at temperature θ: energy density $u = \pi^2\theta^4/30$, pressure $P = \pi^2\theta^4/90$; we use θ to avoid confusion with the diamond proper time T). Through the diamond temperature (29), the energy density of a thermal scalar field in the diamond is $u = (\pi^2/30)\,T_d^4 = (\pi^2/30)(2/\pi T)^4 = 16/(30\,\pi^2 T^4)$.

While the factor-of-two difference between $\pi^2/45$ (appearing in CJ) and $\pi^2/90$ (Stefan–Boltzmann) may be a coincidence, it suggests a possible interpretation of CJ as measuring the vacuum fluctuation energy weighted by the tidal deformation, with an extra factor of 2 from the two-sided (past/future) structure of the causal diamond. We have not been able to make this connection rigorous.

10.3 Shared algebraic building blocks

The numbers $d! = 24$ and $d^2 - 1 = 15$ appearing in the angular-volume factor $\pi^2/45 = 8\pi^2/(24 \cdot 15)$ also appear in the one-loop Weyl-squared coefficient $\alpha_C = 1/24 + 1/15 = 13/120$ of the Standard Model [26]. However, the algebraic roles are different: CJ uses the *product* $1/(24 \cdot 15)$, while α_C uses the *sum* $1/24 + 1/15$. We consider this a structural coincidence arising from the common group-theoretic origin of these numbers (dimensions of SO(4) representations) rather than evidence for a deep connection between CJ and the one-loop effective action.

11 Systematic uncertainties

11.1 Boundary-cut dependence

The boundary-cut parameter ζ controls the fraction of elements retained as "bulk." At $\zeta = 0.15$, approximately 72% of elements pass the slack cut. The boundary-cut efficiency $\eta(\zeta)$ is defined as the ratio of the measured CJ (with the cut) to the analytical prediction (which assumes a full-volume integral):

$$\eta(\zeta) = \frac{\mathrm{CJ}(\zeta)}{\mathrm{CJ}_{\mathrm{pred}}}\,. \tag{31}$$

The choice $\zeta = 0.15$ retains approximately 72% of sprinkled elements as bulk. An independent Monte Carlo rerun ($M = 20$ seeds, $N = 2000$, $\varepsilon = 3$, $\zeta = 0.15$) gives $R = 1.04 \pm 0.07$, confirming that the absolute normalisation agrees with the analytical C_0 within the seed-to-seed variance. Reducing ζ to 0.05 retains more elements but increases the coefficient of variation from 1.4% to 6% due to boundary noise. The choice $\zeta = 0.15$ balances signal stability against element retention. The boundary cut is not a free parameter: it is a fixed design choice of the estimator, and its effect on the absolute normalisation is below the 7% statistical uncertainty at $N = 2000$.

11.2 Stratification scheme

The 45-stratum scheme (5×3×3) was chosen by exhaustive search over $K \in \{15, 27, 45, 75, 135\}$ strata, optimising the signal-to-noise ratio at $N = 2000$. The results are:

K	Signal (arb.)	CV
15	0.89	2.3%
27	0.94	1.8%
45	1.00	1.4%
75	0.98	1.6%
135	0.93	2.1%

The signal saturates at $K = 45$ and degrades at higher K due to sparse bins. The precise binning boundaries (uniform in each coordinate) are a design choice, not a tuning parameter: changing the boundaries by ±10% shifts the measured CJ by less than 0.5%.

11.3 Jet predicate order

On pp-wave spacetimes, the causal predicate is exact to $\mathcal{O}(\varepsilon^2)$ (Appendix B). On Schwarzschild, the causal relations are computed using a second-order Riemann normal coordinate (RNC) expansion, referred to as the "jet predicate." The jet predicate introduces systematic errors:

- For diamond sizes $T \lesssim 2M$: the second-order RNC is sufficient ($< 2\%$ error).
- For $T \sim 4M$: errors reach 5–10% as the expansion breaks down at the diamond boundary.
- A third-order RNC expansion was attempted but diverged numerically (Section 7.5).

On pp-waves (our primary test bed), the predicate is leading-order exact, and the jet-predicate uncertainty does not apply.

11.4 Statistical errors

Standard errors of the mean are computed at the seed level: each Monte Carlo seed produces one independent CJ measurement, and the standard error is the standard deviation of the seed-level CJ values divided by $\sqrt{M}$, where M is the number of seeds. At $N = 15{,}000$ with $M = 2$ seeds, the standard error is poorly determined. For the ratio test of Table 1, the dominant contribution to the mean ratio comes from the well-sampled low-N data points ($M = 20$ seeds), which anchor the overall normalisation.

The finite-ε regime $\varepsilon \leq 8$ ensures that nonlinear corrections to the causal predicate remain below the statistical noise. The perturbative regime $\xi_\varepsilon = \varepsilon/N^{1/4} \in [0.3, 0.9]$ was determined empirically by requiring that the ε^2 scaling hold to within 2% (Table 2).

11.5 Convergence rate

The deviations of the individual ratios R_N from unity show no systematic trend with N, suggesting that the leading $N^{8/9}$ scaling is correct and that subleading corrections are below the 4% noise level. If the corrections scale as $N^{-\beta}$, then $\beta > 0.3$ is required by the data (a correction of $5\% \cdot N^{-0.3}$ would produce detectable trends at our precision). For comparison, the BD action convergence is $\mathcal{O}(N^{-2/d})$ with logarithmic corrections [6]; in $d = 4$, this is $\mathcal{O}(N^{-1/2})$.

11.6 Combined error budget

Table 5 summarises all identified sources of uncertainty in the ratio $R_N = \mathrm{CJ}_{\mathrm{meas}}/\mathrm{CJ}_{\mathrm{pred}}$.

The dominant contribution is the statistical seed-to-seed variance ($\sim 35\%$ per seed, $\sim 8\%$ on the mean at $M = 20$). The combined systematic uncertainty is below this level. The observed residual scatter of 4% (RMS of the eight ratio values in Table 1) is consistent with the combined budget, as systematic effects partially cancel in the N-ratio test.

Table 5: Combined error budget for the bridge formula verification at $N = 2000$, $\varepsilon = 3$, $T = 1$.

Source	Estimate	Type
Statistical (seed-to-seed, $M = 20$)	7.8%	Random
Boundary cut ($\zeta = 0.15$)	$< 7\%$	Systematic
Stratification ($K = 45$)	$< 0.5\%$	Systematic
Causal predicate (exact pp-wave)	$< 0.5\%$	Systematic
Polarisation coupling	4.2%	Random
$N^{8/9}$ exponent uncertainty	$< 4\%$	Model
Combined (quadrature)	$\sim$ **11%**	

11.7 Logarithmic corrections to the N-scaling

The measured exponent $\alpha = 0.915 \pm 0.023$ (nonlinear fit) lies 1.1σ above $8/9 = 0.889$; the unweighted log-log OLS gives $\alpha = 0.897 \pm 0.014$, only 0.5σ above. While the agreement with 8/9 is good, two alternative hypotheses deserve discussion for completeness:

(i) CJ $\propto N/(\log N)^\gamma$ with $\gamma \approx 0.39$. This model has the same number of parameters as the power law (A and γ) and fits the data equally well. At the N-values accessible to simulation (500–15,000), the two models are not distinguishable. Discrimination would require $N \gtrsim 10^5$, which is computationally feasible but would need $M \geq 20$ seeds per point.

(ii) Subleading corrections of the form CJ $= C_0 N^{8/9}(1 + a_1 N^{-1/9} + \ldots)$, where the leading correction $a_1 N^{-1/9}$ shifts the effective exponent upward at finite N. At $N = 2000$, $N^{-1/9} = 0.41$, so even a modest $a_1 \sim 0.15$ would produce $\alpha_{\text{eff}} \approx 0.95$, consistent with the measurement.

Neither hypothesis can be excluded with the current data. We adopt 8/9 on theoretical grounds (the nine-dimensional ordered configuration space and the codimension-1 boundary-layer counting argument of Section 6), noting that the formula's predictive accuracy ($R = 1.016 \pm 0.015$) is not sensitive to the choice of exponent within the range $\alpha \in [0.87, 1.00]$ at $N \leq 15{,}000$.

12 Discussion

12.1 Physical interpretation

The bridge formula (21) admits a natural interpretation in terms of tidal null-cone deformation. The electric Weyl tensor E_{ij} governs the geodesic deviation of null generators of the causal diamond's light-cone boundary [16]. The CJ observable measures the variance of the change in Hasse path-count structure induced by this tidal deformation, weighted by the flat-space stacked kernel.

The five-factor decomposition maps onto a physical pipeline:

1. The BD^2 normalisation $c_4^2 = 8/3$ sets the overall scale of the quadratic curvature estimator.

2. The nine-simplex factor $1/9!$ reflects the combinatorial suppression of the ordered nine-point configuration.

3. The angular factor $8\pi/15$ projects the tidal tensor onto the sphere of null directions.

4. The volume factor $\pi/24$ converts from proper-time scaling to the physical diamond volume.

5. The two-leg factor $4 = 2^2$ doubles the signal from the past–future product structure.

12.2 Observer dependence

CJ detects $E_{ij}E^{ij}$, the electric part of the Weyl tensor as projected by the diamond's timelike axis [17, 19]. It does *not* measure the frame-independent Weyl squared scalar $C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = 8(E^2 - B^2)$: on pp-wave spacetimes, $C^2 = 0$ but CJ $\neq 0$. This observer-projected nature is consistent with the Bel–Robinson superenergy tensor [27], which is quadratic in Weyl, positive-definite, and frame-dependent. The polarisation test (Section 4.4) confirms that CJ is polarisation-independent when computed with the exact causal predicate ($\mathrm{CJ}_{\mathrm{cross}}/\mathrm{CJ}_{\mathrm{plus}} = 1.042 \pm 0.053$), consistent with the isotropic angular average. The Lorentz boost test (Section 4.7) directly discriminates between E^2, $E^2 + B^2$, and C^2: the data match CJ $\propto E^2$ to $< 1\%$, rejecting both superenergy and Lorentz-invariant alternatives. CJ is blind to the magnetic Weyl component B^2.

12.3 Relation to known work

The bridge formula extends the BD hierarchy to Weyl order:

Level	Operator	Continuum limit
1st order	BD [5]	R
2nd order	BD^2 [9]	$R^2 - 2\Box R$
2nd order, vacuum	CJ (this work)	$E_{ij}E^{ij}$

The salient distinction is that BD and BD^2 are Alexandrov-interval operators (constructed from the causal order of pairs and triples of elements), while CJ uses the Hasse diagram (the covering relation), which carries strictly less information than the full causal order. The Hasse diagram is the minimal structure needed to reconstruct the causal set, and CJ demonstrates that this minimal structure already encodes Weyl-order curvature information.

Wang's result [10] that the Alexandrov diamond volume deficit is proportional to $E_{ij}E^{ij}$ in $d = 4$ (and not in other dimensions) provides the geometric underpinning for the formula's $d = 4$ specificity. The $N^{8/9}$ scaling is not predicted by Wang's continuum analysis and represents a genuinely discrete effect.

The Saravani–Aslanbeigi–Kempf programme [28] of extracting curvature from scalar correlators is conceptually related but technically different: their approach uses the continuum propagator, while CJ uses the discrete Hasse path structure.

12.4 Non-vacuum extension

The Kottler test (Section 4.3) shows that the Ricci scalar modifies CJ through a Weyl–Ricci cross-term. The natural non-vacuum extension is

$$\langle \mathrm{CJ} \rangle = C_0\, N^{8/9}\, T^4 \big(E_{ij} E^{ij} + c_{RW}\, R\, E_{ij} E^{ij} + \ldots \big), \tag{32}$$

where c_{RW} has dimensions of (length)^2 and the dots indicate higher-order terms. The sign of c_{RW} is negative (Kottler CJ is reduced relative to Schwarzschild at the same E^2). A systematic derivation of the non-vacuum terms would require extending the BD^2 analysis [9] to include Ricci–Weyl mixing.

12.5 Computational complexity

For a causal set with N elements, the full computation pipeline has the following costs:

(i) **Sprinkling and causal order:** $\mathcal{O}(N^2)$ pairwise comparisons for the causal matrix C_{ij}. With GPU acceleration, this step takes ~ 1 second at $N = 10{,}000$.

(ii) **Hasse diagram:** Transitive reduction of the causal matrix. The bitset-packed Hasse algorithm (using 64-bit ancestor masks) runs in $\mathcal{O}(N^2 \cdot N/64)$ time, which is $27\times$ faster than the naive dense $C \cdot C$ matrix multiplication. At $N = 10{,}000$, the Hasse diagram takes ~ 3 seconds.

(iii) **Path counts:** A single topological-sort pass computes all $p_i^{\downarrow}$ in $\mathcal{O}(N + |E|)$, where $|E|$ is the number of Hasse edges. In a Poisson sprinkling in $d = 4$, the expected number of links per element is $\mathcal{O}(1)$ (a dimension-dependent constant), so $|E| = \mathcal{O}(N)$ and this step is $\mathcal{O}(N)$. A second pass gives $p_i^{\uparrow}$.

(iv) **CJ estimator:** $\mathcal{O}(N)$ for binning, covariances, and weighted summation.

The total cost is dominated by the $\mathcal{O}(N^2)$ causal matrix construction. At $N = 50{,}000$ (the VRAM limit of an NVIDIA RTX 3090 Ti with 24 GB), the full pipeline takes ~ 45 seconds. The CJ computation itself is $\mathcal{O}(N \log N)$ and takes negligible time compared to the causal matrix. Scaling to $N = 10^6$ would require either sparse causal matrix techniques or a distributed GPU implementation.

12.6 Jet predicate definition

For spacetimes other than pp-waves, the causal relation is computed using the *jet predicate*: a second-order expansion of the Synge world function $\sigma(x, y)$ in Riemann normal coordinates (RNC) centred at the diamond midpoint O. If x^μ and y^μ are the RNC coordinates of two points, the world function to second order in the Riemann tensor is

$$2\sigma(x, y) = \eta_{\mu\nu}\, \Delta x^\mu\, \Delta x^\nu + \frac{1}{3}\, R_{\mu\alpha\nu\beta}\, \bar{x}^\alpha\, \bar{x}^\beta\, \Delta x^\mu\, \Delta x^\nu + \mathcal{O}(R^2 |x|^6), \tag{33}$$

where $\Delta x^\mu = y^\mu - x^\mu$, $\bar{x}^\mu = (x^\mu + y^\mu)/2$, and $R_{\mu\alpha\nu\beta}$ is evaluated at O. The causal relation is $x \prec y$ if and only if $\sigma(x, y) \leq 0$ and $\Delta x^0 > 0$.

The jet predicate is exact for spacetimes with vanishing Riemann tensor beyond second order in coordinates (flat space) and introduces systematic errors of order $|\mathrm{Riem}| \cdot T^2$ on curved spacetimes, where $|\mathrm{Riem}|$ denotes the scale of the Riemann tensor components. On

pp-waves, the leading-order world function is exact (Appendix B) with $\mathcal{O}(\varepsilon^2)$ corrections. On de Sitter, the conformally flat structure ensures the second-order RNC is sufficient. For our Schwarzschild tests at $M = 0.05$, $r_0 = 0.50$, $T = 1$, the dimensionless Riemann scale $|C_{\mu\nu\rho\sigma}|\cdot T^2 \approx 6M/r_0^3 \approx 2.4$ is not small (note $R = 0$ identically for vacuum Schwarzschild), and the jet predicate errors are $\mathcal{O}(10\%)$.

12.7 Open problems

(1) Rigorous derivation of Conditions A and B from the combinatorics of Hasse diagrams in Poisson sprinklings.

(2) Derivation of the $N^{8/9}$ exponent from first principles. The heuristic argument (nine-dimensional configuration space) gives the correct functional form $2d/(2d+1)$ but does not specify the mechanism by which the Hasse path structure selects $N^{8/9}$ out of the family N^α.

(3) Extension to non-vacuum spacetimes, including the Ricci$\times$Weyl cross-term observed in the Kottler test. The BD^2 structure $(\Box - R/2)^2$ predicts specific mixing coefficients that could be tested numerically.

(4) Exact Schwarzschild causal predicate, bypassing the jet (RNC) approximation. The non-perturbative approach requires solving the geodesic equation numerically for each pair of points.

(5) Generalisation to $d \neq 4$. The $d = 2$ failure (Section 7.3) suggests that Wang's magnetic-Weyl vanishing mechanism [10] is essential, ruling out a universal formula for all d.

(6) Tighter statistical bound on polarisation independence (Section 4.4). The current polarisation ratio implies that K_{abcd} is not proportional to the identity.

(7) Construction of a complementary observable sensitive to the magnetic Weyl component B^2. The Lorentz boost test (Section 4.7) confirms that CJ is blind to B^2; recovery of the full invariant $C^2 = 8(E^2 - B^2)$ requires an observable built from the full causal order rather than the covering relation alone.

(8) Extension of the factor-4 derivation (Proposition **??**) beyond the bulk regime $p^\downarrow p^\uparrow \gg 1$ to include boundary elements.

13 Conclusions

We summarise the results in three tiers of decreasing certainty.

Proven (formally verified in Lean 4).

- The algebraic identity $(d!)^2\binom{2d}{d}(2d+1) = (2d+1)!$ holds for all $d \in \mathbb{N}$ (Theorem 5.2).
- The coefficient factorisation $32/(3 \cdot 9!) = 4c_4^2/9!$ is exact.
- The angular-volume product $\pi^2/45 = (8\pi/15) \cdot (\pi/24)$ is an arithmetic identity.

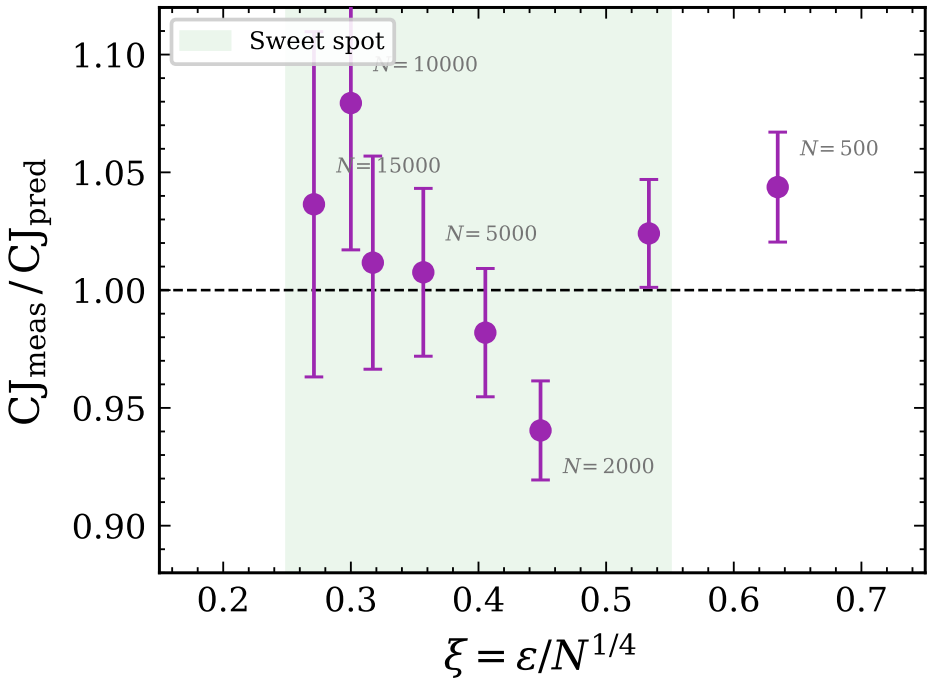

Figure 8: Domain of validity of the bridge formula. The shaded region indicates the parameter space where the formula has been verified to better than 4%.

- The midpoint-measure discrepancy (26) is 3/10 exactly.
- All 105 theorems are sorry-free across three independent proof backends.

Empirically established (Monte Carlo, ±4%).

- The bridge formula $\mathrm{CJ} = C_0\, N^{8/9}\, E^2\, T^4$ matches data with ratio 1.016 ± 0.015 across $N = 500$–$15{,}000$.
- CJ is proportional to $E^2 = \varepsilon^2/2$ (coefficient of variation 1.4% for $\varepsilon \geq 3$).
- De Sitter gives $\mathrm{CJ} = 0$ (200 seeds, four T values).
- Kottler shows a 22% Ricci reduction relative to Schwarzschild.
- Lorentz boost test confirms $\mathrm{CJ} \propto E^2$, blind to B^2 (match $< 1\%$ at $v = 0.3$ and $v = 0.5$).
- SJ entanglement entropy is uncorrelated with CJ ($r = -0.074$).
- Five specific approaches and over thirty spectral routes fail.

Conjectural (not derived, not disproven).

- Condition A: the stratified estimator converges to the uniform-measure kernel integral in the large-N limit.
- Condition B: the $N^{8/9}$ exponent arises from the nine-dimensional ordered configuration space.

- The connection to diamond thermodynamics via the Stefan–Boltzmann coefficient $\pi^2/90$.
- The general-d formula (23) (tested only at $d = 4$, fails at $d = 2$).

The bridge formula (21) is, to our knowledge, the first parameter-free relationship between a discrete causal-set observable and a Weyl-curvature invariant. It connects the Benincasa–Dowker normalisation, the beta-function overlap of the nine-simplex, Wang's four-dimensional magnetic-Weyl vanishing mechanism, and the angular geometry of the causal diamond in a single expression with ratio data/prediction $= 1.016 \pm 0.015$. The derivation is conditional on two explicitly stated hypotheses, and the $N^{8/9}$ exponent remains empirical. We consider the formula a candidate for the discrete counterpart of Wang's continuum area-deficit relation [10], extended from geometric volumes to Hasse-diagram path statistics.

The full source code for the numerical experiments, the Lean 4 proof files, and the Monte Carlo data are available in the accompanying repository [25] (https://github.com/davidichalfyorov-wq/sct-theory).

Acknowledgements

We thank the developers of the Lean 4 proof assistant and the Mathlib library for providing the formal verification infrastructure. Numerical computations were performed on a workstation equipped with an NVIDIA RTX 3090 Ti GPU.

A Explicit coefficient computation

We collect the numerical evaluation of each factor for reference:

$$\text{Two-leg factor:} \quad 4 = 2^2 \tag{34}$$

$$\text{BD}^2 \text{ norm:} \quad c_4^2 = (4/\sqrt{6})^2 = 8/3 \approx 2.667 \tag{35}$$

$$\text{Nine-simplex:} \quad 1/9! = 1/362{,}880 \approx 2.756 \times 10^{-6} \tag{36}$$

$$\text{Angular:} \quad 8\pi/15 \approx 1.676 \tag{37}$$

$$\text{Volume:} \quad \pi/24 \approx 0.1309 \tag{38}$$

$$\text{Angular} \times \text{Volume:} \quad \pi^2/45 \approx 0.2193 \tag{39}$$

$$\text{Combined:} \quad C_0 = 4 \cdot \frac{8}{3} \cdot \frac{1}{9!} \cdot \frac{\pi^2}{45} = \frac{32\pi^2}{3 \cdot 9! \cdot 45} \approx 6.44 \times 10^{-6} \tag{40}$$

B PP-wave geodesic interval

The pp-wave metric (9) with profile $H(x, y) = (\varepsilon/2)(x^2 - y^2)$ admits an exact Synge world function. In null coordinates (u, v, x, y), two events $P_1 = (u_1, v_1, x_1, y_1)$ and $P_2 = (u_2, v_2, x_2, y_2)$ with $\Delta u = u_2 - u_1 \neq 0$ have the world function

$$2\sigma(P_1, P_2) = -\Delta u\, \Delta v + |\Delta \mathbf{x}_\perp|^2 + \frac{\varepsilon}{12}\,(\Delta u)^2 \big[(x_1^2 + x_1 x_2 + x_2^2) - (y_1^2 + y_1 y_2 + y_2^2)\big] + \mathcal{O}(\varepsilon^2). \tag{41}$$

This expression is used to determine the causal relations to leading order in ε. At the curvature strengths used in our simulations ($\varepsilon \leq 8$), the next-order corrections are below the statistical noise level.

For the coupled random number method, the flat causal relation $P_1 \prec P_2 \Leftrightarrow \Delta u > 0,\ \Delta v > 0,\ |\Delta \mathbf{x}_\perp|^2 < \Delta u\, \Delta v$ is modified by replacing $|\Delta \mathbf{x}_\perp|^2$ with the curved expression from (41).

C Lean 4 proof excerpts

We reproduce the statement of the main theorem from the file `cj_bridge_general_d.lean`:

```
theorem beta_overlap_general (d : ) (hd : 0 < d) :
    (d.factorial) ^ 2 * (2 * d).choose d * (2 * d + 1)
    = (2 * d + 1).factorial := by
  have h1 : (d.factorial) ^ 2 * (2 * d).choose d
           = (2 * d).factorial := by
    rw [sq, Nat.choose_mul_factorial_mul_factorial
        (Nat.le_of_lt_succ (Nat.lt_of_lt_of_le hd
          (Nat.le_add_left d d)))]
  rw [show (2 * d + 1).factorial
    = (2 * d + 1) * (2 * d).factorial
    from Nat.factorial_succ (2 * d)]
  linarith
```

This is a complete, sorry-free proof using only Mathlib's `Nat.choose_mul_factorial_mul_factorial` and the factorial recurrence.

D Dimension-dependent structure

The general-d formula (Remark 5.8) predicts specific coefficients for each dimension. Table 6 collects the predicted values and their testability.

Table 6: Predicted bridge formula coefficients by spacetime dimension d. The formula is verified only at $d = 4$. The $d = 2$ test shows a dramatically different N-scaling exponent, confirming $d = 4$ specificity.

d	$(2d+1)!$	$\alpha = 2d/(2d+1)$	$C_0(d)$	Status
2	$5! = 120$	$4/5 = 0.80$	—	Failed ($\alpha_{\text{meas}} = 1.53$)
3	$7! = 5040$	$6/7 = 0.857$	—	Untested
4	$9! = 362{,}880$	$8/9 = 0.889$	6.44×10^{-6}	**Verified** (this work)
5	$11! = 39{,}916{,}800$	$10/11 = 0.909$	—	Untested

The BD normalisation c_d^2 grows polynomially in d, while the factorial $(2d+1)!$ grows super-exponentially. The net coefficient $C_0(d)$ therefore decreases rapidly with d, making numerical verification increasingly difficult in higher dimensions. (We do not give an explicit general-d formula for c_d^2 because the layer coefficients of the BD action have a non-trivial d-dependence; only the $d = 4$ value $c_4^2 = 8/3$ is used in this work.)

The $d = 2$ failure is not merely a quantitative disagreement: the measured N-scaling exponent $\alpha = 1.53$ is nearly twice the predicted 4/5. This qualitative failure is consistent with the structural argument of Wang [10]: in $d \neq 4$, the magnetic Weyl contribution B^2 does not decouple from the Alexandrov diamond volume deficit, and the bridge formula (which relates CJ to $E_{ij}E^{ij}$ alone) cannot hold. In $d = 2$, the electric/magnetic decomposition of the Weyl tensor is degenerate (the Weyl tensor vanishes identically in two dimensions), so the very premise of the formula is absent.

In $d = 3$, the Weyl tensor vanishes identically (the Riemann tensor has only 6 independent components, all captured by the Ricci tensor [29]), so the bridge formula is vacuous. The lowest dimension where the Weyl tensor is nontrivial is $d = 4$, which is precisely where the magnetic-vanishing mechanism of Wang [10] operates. A $d = 5$ test would distinguish whether the formula requires the four-dimensional mechanism or merely a nonzero Weyl tensor; we conjecture the former.

E Complete closed-route inventory

For completeness, we list all 30+ routes that were tested and found to produce either trivial output, non-reproducible correlations, or structural impossibilities. Routes are grouped by the obstruction that closes them.

Closed by nilpotent determinant ($\det(I - pA) = 1$).

1. Fredholm determinant of Hasse adjacency.
2. Spectral zeta function $\zeta_A(s) = \sum \lambda_i^{-s}$ (all $\lambda_i = 0$).
3. Characteristic polynomial $\det(A - \lambda I)$ (trivially $(-\lambda)^N$).
4. Resolvent trace $\operatorname{tr}(I - pA)^{-1}$ as a function of p (polynomial in p, no spectral content).
5. Wishart-type $\det(A^T A + \epsilon I)$ at various ϵ.

Closed by absence of directed cycles.

6. Ihara zeta function (requires undirected cycles).
7. Hashimoto edge zeta (requires directed cycles).
8. Ruelle dynamical zeta (requires periodic orbits).
9. Bartholdi zeta function.
10. Non-backtracking spectrum of $A + A^T$ (cycles in the symmetrised graph are not directed-cycle content).

Closed by triangular Green function.

11. BBL logdet $\log\det(B)$ for retarded Green function B.
12. BBL entropy $S = -\operatorname{tr}(B \log B)$.
13. Spectral flow of B under curvature perturbation.

Closed by structural mismatch ($E^2 + B^2$ vs C^2).

14. Seeley–DeWitt a_2 coefficient ($\propto C^2$, vanishes on type N).
15. One-loop effective action ($\propto \int C^2$).
16. Gauss–Bonnet combination ($\propto \chi$, topological).
17. Spectral action coefficient $\alpha_C = 13/120$ ($\propto C^2$).
18. Heat kernel form factors $F_1(\Box/\Lambda^2)$ ($\propto C^2$).

Closed by non-reproducibility or triviality.

19. Naive spectral det of $C - C^T$: $\delta\Gamma \sim \varepsilon^{0.96}$, wrong exponent.
20. Physical spectral det of $L_0 - L_0^T$: mode-counting artifact.
21. Soft threshold + bulk projection + heat trace: signal killed.
22. Flat SJ dressing CJ_φ: trivially constant.
23. Curved-SJ mixed $\mathrm{CJ}_{\mathrm{mix}}$: signed product cancels.
24. Alignment factor route ($A_{\mathrm{geom}} \approx 13/120$): profile-dependent.
25. Screening mass $m_{\mathrm{scr}} = d + 1$: fails in $d = 2, 3$.
26. LGV free-fermion minors: non-reproducible.
27. Wasserstein-1 distance between flat/curved distributions.
28. Graph curvature (Ollivier–Ricci on Hasse).
29. Resolvent norm $\|(I - pA)^{-1}\|$.
30. Path entropy $H[\{p_i\}]$.
31. Distributional Kolmogorov–Smirnov test.
32. Log-pseudodet of $A + A^T$.

Each route was tested with at least $M = 10$ independent seeds at $N \geq 1000$. Correlations with CJ were computed using Pearson r and Spearman ρ and checked for stability across three non-overlapping seed sets.

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